



Imaging Insights

User Manual

Software Version 2.2.3

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Revision 2
US English

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User Manual

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Revision 2

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Manufacturer:

GE Medical Systems, operating as GE HealthCare

2 rue Marie Hamm

67000 Strasbourg

FRANCE

Revision History

Revision	Date	Reason for Change
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1 Legal Information

1.1 Copyright Notice

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1.4 Publication Disclaimers

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The Imaging Insights User Manual is an independent publication and has not been authorized, sponsored, or otherwise approved by Apple Inc.

2 About this Manual

2.1 Intended Audience

The Imaging Insights User Manual is intended for users and consumers of Diagnostic Imaging data, such as Radiology Directors, Department or Service Line Managers, COOs, Process Improvement personnel, and in some cases Technologists and Medical Physicists.

2.2 User Interface Note

The screen captures in this document are intended for demonstration purposes only and may not always be fully identical to the actual user interface (e.g., in color scheme). Use these screen captures as guides.

2.3 Demographic Information Disclaimer

Any patient identifiable information contained in this manual, including but not limited to name, age, gender, date of birth, medical record number and exam date is fictitious information and was generated for the purposes of illustration only.

3 About Imaging Insights

3.1 Product Description

Imaging Insights dashboards deliver multi-vendor and multi-modality Utilization, Standardization, Protocol and Dose analytics dashboards. These dashboards help users to identify improvements opportunities for clinical operations improvements, such as increasing patient throughput, standardization of care or radiation dose management. Practice level metrics are provided (excluding patient-specific information) to help users to identify operational and clinical improvements and to track changes over time.

3.2 Browser Compatibility

The following browsers are used to run Imaging Insights:

- Computers with Windows® operating system
 - Microsoft Edge® and Internet Explorer® 11



NOTE

Compatibility mode for Internet Explorer® is not supported.

- Google Chrome™.
- Computers with macOS® operating system
 - Google Chrome™.
 - Firefox® and Firefox ESR
 - Safari®
- Google Chrome™ on mobile devices with Android™ operating system
- Safari® on mobile devices with iOS 11.3 (or later) operating system.



NOTE

Google Chrome™ is the preferred browser

4 Getting Started

4.1 Signing in and out of the Application

Use this procedure to sign in and out of Imaging Insights.

1. Start the application in your web browser. The Imaging Insights sign-in page appears.
2. In the **Username** field, enter/type your username registered with your account.
3. Click **Continue**.
4. In the **Password** field, type your password.
5. Click **Sign In**.



NOTE

If you have entered your username and/or password incorrectly or your user account has been deleted since your latest session for any reason, the application displays the following message:

`"Invalid username or password"`

- a. Retype your username and password.
- b. Click **Sign In**.

If the problem persists, but you are confident that you are using the correct username and password, contact your administrator for assistance.

6. For MFA choose Email or Mobile App.
 - a. Enter the code received based upon your choice above. Note the code is sent to the registered email address or phone number.
 - b. Click **Authenticate**.
7. To sign out of the application:
 - a. On the application banner, click the name drop-down menu to display the menu options.
 - b. Click **Sign Out**.

The Imaging Insights sign-in page appears.

4.2 Accessing the Imaging Insights Modality Module Dashboards

Use this procedure to access Imaging Insights Modality Module dashboards.

1. Sign in to Imaging Insights. For additional information, see [4.1 Signing in and out of the Application on page 8](#).
2. From the navigation pane on the left, click **Explore**. The project list appears on the right.
3. From the project list, click **Imaging Insights**. The contents of the Imaging Insights project appears.

4. From the Imaging Insights, select the module of your interest, the contents of the selected Imaging Insights module appears.
5. From the selected Imaging Insights module, click **Home**. The Home dashboard of the selected Imaging Insights module appears. You can navigate between the dashboards as described in [5.3 Navigation in Imaging Insights on page 11](#).

5 Reviewing Dashboards

5.1 Modalities Dashboards

For information about accessing the dashboards, see [4.2 Accessing the Imaging Insights Modality Module Dashboards on page 8](#).

Following modalities are covered: Computed Tomography (CT), Mammography (MG) and Magnetic Resonance Imaging (MR), and Ultrasound (ULS). Depending on the modalities not all dashboards are provided.

The following dashboards are available in the Imaging Insights Module:

- **Home**
 - Available for all modalities
 - The widgets on the Home dashboards provide basic indicators on device utilization, protocol standardization and quality of data from devices to the application.
- **Utilization**
 - Available for all modalities
 - The widgets on the Utilization dashboards helps you to understand your imaging volume. The widgets provide data about diagnostic imaging exams: where, when, on what systems, and by which technologists they were carried out. This information can help you to identify opportunities for clinical operations improvements, such as increasing patient throughput.
- **Standardization**
 - Not available for Ultrasound
 - The widgets on the Standardization dashboards help you to find out how many of the protocols mapped to Standard Study Descriptions were performed in a standard way, and to identify the important sources of variation.
- **Protocol**
 - Available only for the CT and MR modalities
 - The widgets on Protocol dashboards help you to find out how consistently imaging exams were carried out by providing data by devices, study types and technologists, and to identify variation in imaging techniques.
- **Dose**

The Dose dashboard is available only for the CT and MG modalities.

 - The widgets on the Dose dashboard help you to baseline dose performance, to detect when and in which patient populations dose outliers are happening, and to identify the most important factors contributing to high dose levels.



NOTE

For all modalities except Ultrasound Data preceding the date of accessing the Imaging Insights (for up to 18 months) is available only if there has been DoseWatch™ connection.

5.2 Dose Analytics Dashboards

The Dose Analytics provides interactive dashboards from modalities that deliver dose to patients. For more details on Dose Analytics dashboards and associated capabilities see the dedicated Help dashboard as well as Annex A and B sections of this document.

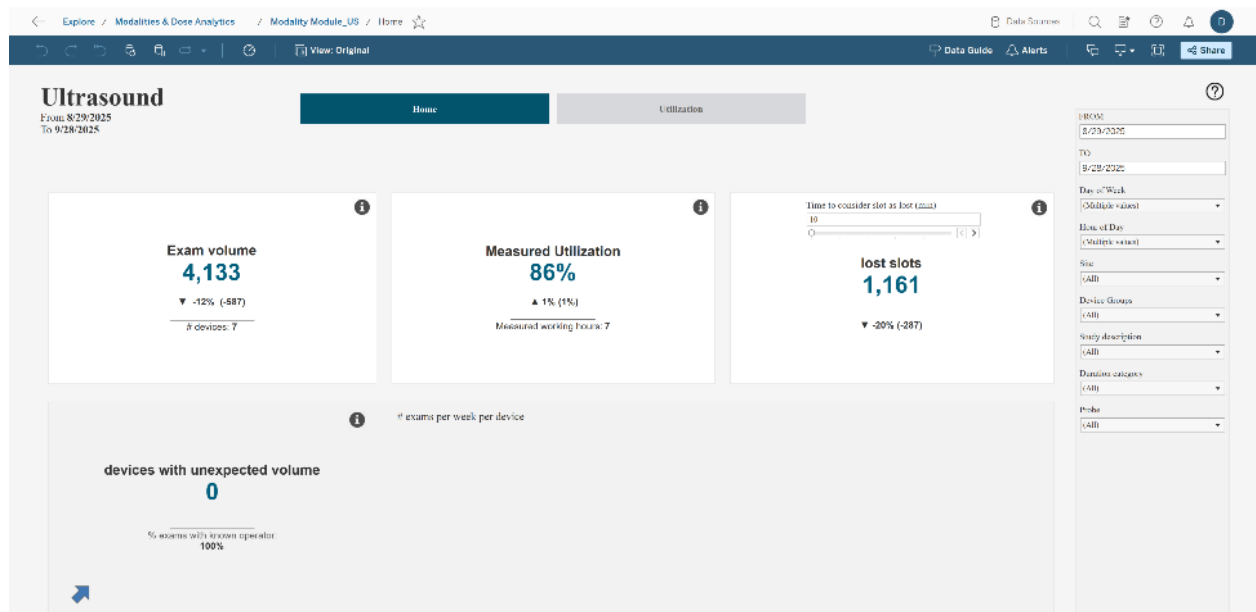


NOTE

A DoseWatch™ connection is mandatory. Dashboards can be accessed through DoseWatch™.

5.3 Navigation in Imaging Insights

Once you have opened a dashboard, you can navigate between the dashboards as follows.



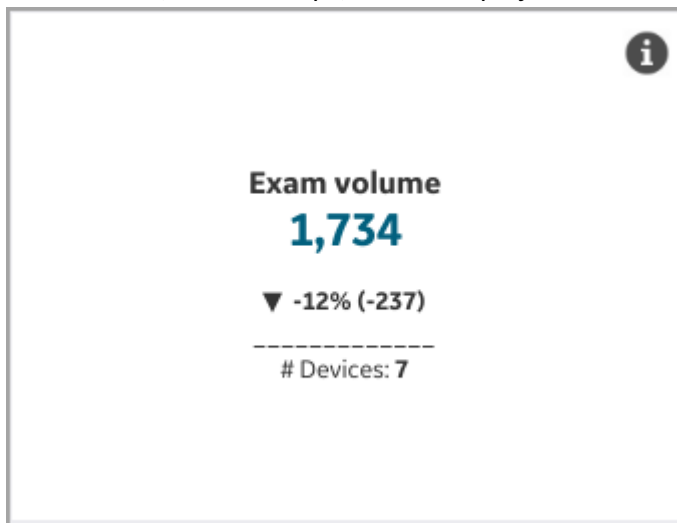
1. Navigate within the application/between the modalities using the breadcrumbs on the application banner.
For example, if you are on the Utilization dashboard for **CT** and you want to review the **MR** dashboards, click **Imaging Insights**, and then select Module **MR**.
2. Navigate between the dashboards for the selected modality using the dashboard tiles.
Also, using the "navigate to" and/or "open" buttons, you can access additional data charts and tables (e.g., exam details and input data tables). For detailed information, see the following sections:

5.4 Actions Available from the Widgets/Application

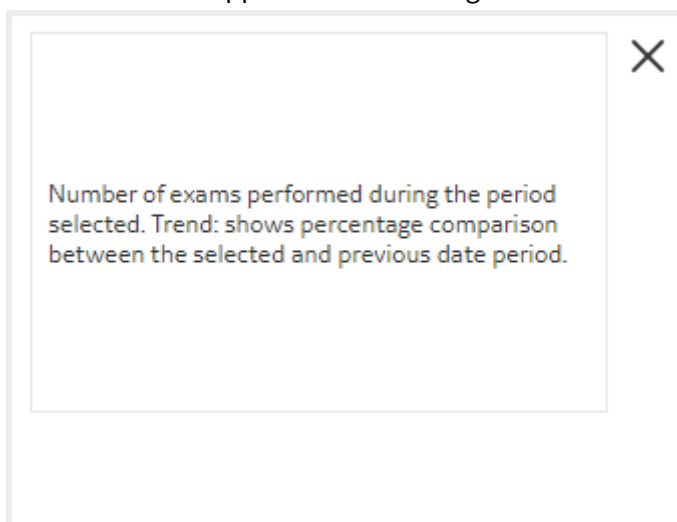
The following actions are available from the widgets/application:

Displaying infotips

- Click the  (Show 'Infotip") icon to display additional information about the selected widget.

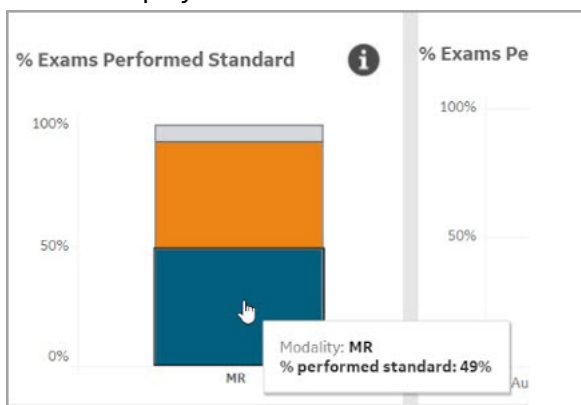


- The information appears over the widget.

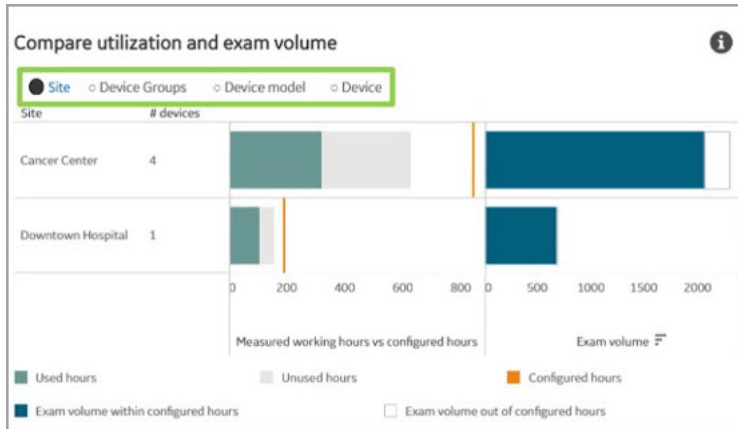


Click the X icon to close the infotip.

- Hover the pointer over a column, column color, slice, point, and so on in a chart or a value in a table to display additional details.

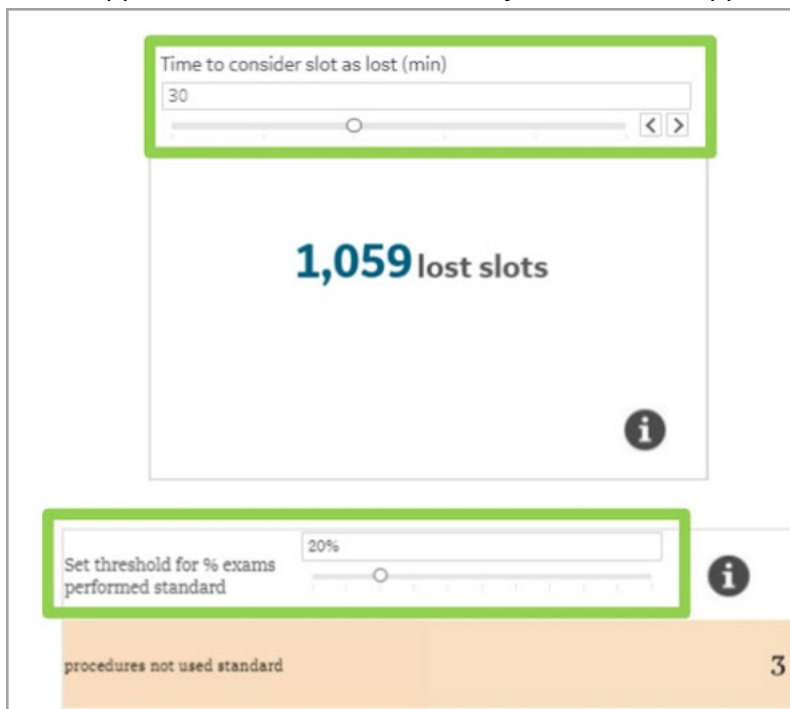


- Where applicable, use the "show-by" options to select the category you want to view the corresponding data.



This selection applies only to the corresponding charts.

- Where applicable, use the slider to modify the threshold applied in the calculation of the widget.



- Click a column, column color, slice, point, and so on in a chart or a value in a table to display information for the selected data item. All other widgets on the dashboard are updated accordingly, showing data only for the selected data item. Click the selected column, column color, slice, point, and so on in a chart or a value in a table again to deselect it and display the data for the complete chart/table on the dashboard.
- From widgets where the  (Zoom in) icon is available, click the icon to open the widget in full view mode. Click on  (Zoom out) icon to return to the original dashboard.

**NOTE**

The action filters applied on the dashboard will be applicable in the zoomed view. Action filters are available in the zoom view, any changes done to the action filter in zoom view will be translated back to the original dashboard on zoom out.

- Undo, Redo, and Revert to published (or custom view).



From the application toolbar:

- Click the **Undo** (or ) button to undo an action you performed.

If you have applied multiple filters, you can use the **Undo** (or ) button for multiple clicks to remove filters one after another in reverse order (i.e., removing first the most recent filter to the first filter applied).

- Click the Redo (or ) button to redo an action that you previously performed.

- Click the **Revert** (or ) button to revert to the view at the time it was published or to the state the custom view was saved in.

If you have applied multiple filters and you want to remove them all at one stroke, we recommend

using the Revert (or ) button.

For more information about custom views, see [5.6 Creating Custom Views on page 14](#).

5.5 Filtering and Sorting Options

The Imaging Insights provides filtering capabilities to drill down on the information provided. The filtering options are available on the right side of the dashboards. Filtering options selected are retained while navigating through the dashboards. One or combination of multiple filters can be used for filtering information in the dashboard.

To sort data in tables, click the  (sort in alphabetically ascending/descending order) or  (sort in ascending/descending order) button in the corresponding header as applicable, depending on the data type (names, dates, etc.).

5.6 Creating Custom Views

You can create custom views with your selections and filters applied and saved.

Use this procedure to create custom views in the Imaging Insights.

1. Filter the data, change sorting, select specific marks and/or make any other modifications as needed.
2. From the application toolbar, click **View**. The Custom Views dialog box appears.

3. In the **Name this view** filed, type a name for the custom view.
4. Optionally, select the **Make it my default** check box if you want to make the custom view your default view.
5. Optionally, select the **Make visible to others** check box to make the custom view available to other users.
6. Click **Save**
If you have selected the **Make visible to others** check box, click **Share** from the view's toolbar to share the custom view with other users.

**NOTE**

The customer's email server must be configured on the Tableau® server to share custom views.

5.7 Additional Actions Available for Dashboards

5.7.1 Creating alerts

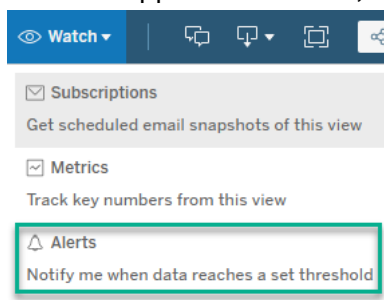
When data reaches important thresholds for your business, data-driven alerts automatically send email notifications to key people you specify. Alerts can be added to charts that have continuous numeric axis.

**NOTE**

The use of Alerts feature is facilitated through email delivery. By default, these features are inactive and may only be activated in accordance with local laws and regulations. For further details, please consult with a representative from GE HealthCare.

Use this procedure to create alerts in Imaging Insights.

- From the application toolbar, click **Alerts**.



The Alerts pane appears on the right.

- Select the continuous numeric axis of a chart.

- From the Alerts pane, click the **Create** button, the Create Alert window appears.

- To Set up the alert set the following:

- **Condition** (drop-down list)
- **Threshold** (text field)
- **Subject** (text field)



NOTE

This field is pre-populated but editable.

- Send email frequency (**When the condition is true, send email** drop-down list).
- **Recipients** field. Start typing the username of a user to add the corresponding user to the alert email recipients.
- **Email** (text field).



NOTE

You must enter valid email address.

- Optionally, select the **Make visible to others** check box. If this check box is selected, everyone with access to the related view can see and add themselves to the alert.
- Click the **Create Alert** button. The alert is added to the Alerts pane. From the Alerts pane, the following actions are available for the alert by clicking the Actions drop-down menu:
 - **Remove me**
 - **Edit**

- **Change owner**
- **Delete**

5.7.2 Subscribing to views/workbooks

Your subscriptions email you an image or PDF snapshot of a view or workbook at regular intervals without requiring you to sign in to Tableau® Server or Tableau Online.

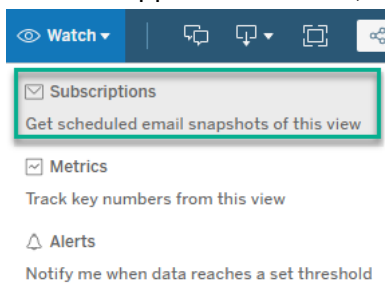


NOTE

The use of Subscription feature is facilitated through email delivery. By default, these features are inactive and may only be activated in accordance with local laws and regulations. For further details, please consult with a representative from GE HealthCare.

Use this procedure to subscribe to views/workbooks in Imaging Insights.

- From the application toolbar, click **Subscribe**.



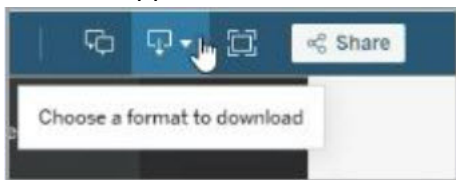
The Subscribe window appears.

- To set up the subscription, Set the following:
 - **Subscribe Users** field
Start typing the name of a user or user group to add the corresponding user/user group to the subscription.
Select the **Subscribe me** check box to add yourself to the subscription.
 - **Include (drop-down list)**
Select **This View** or **Entire Workbook** as applicable.
Select the **Don't send if view is empty** check box if you do not want to receive a snapshot of an empty view. The **Don't send if view is empty** check box is active only if you have selected the **This View** option.
 - **Format** (drop-down list)
Select the format in which you want to receive the snapshot (image, PDF or both).
 - **Subject** (text field)
This field is pre-populated but editable.
 - **Message** (text field, optional)
 - **Schedule** (drop-down list)
Select **Monday morning** or **Weekday mornings** as applicable.
 - Click the **Subscribe** button. The following message appears:
"Success: Created <number of subscriptions> subscriptions." The subscription is added to the Subscriptions tab of the workbook.

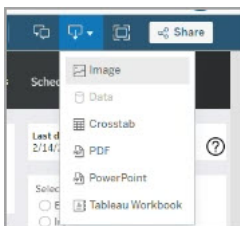
5.7.3 Downloading dashboards

Entire workbooks, dashboards or sheets can be downloaded in different output formats. Use this procedure to download workbooks, dashboards or sheets in the selected format in Imaging Insights.

1. From the application toolbar, click **Download**.



The **Download** window appears.



2. Select the format in which you want to download the selected workbook, dashboard or sheet(s). Select the **Image** option to download the dashboard/sheet in Portable Network Graphics (.PNG) format.
 - a. From the Download window, click **Image**. The **Save/Save As** dialog box appears.

- b. Save the .PNG file to the location of your choice. Select the Data option to download the dashboard/sheet in Comma Separated Values (.CSV) format.
 - i. From the Download window, click **Data**. The View Data window appears.
 - ii. From the View Data window, click the **Download all rows as a text file** link. The **Save/Save As** dialog box appears.
- c. Save the .CSV file to the location of your choice. Select the **Crosstab** option to download a sheet as a text table/cross tabulation in .CSV format.
 - i. From the Download window, click **Crosstab**. The Download Crosstab window appears.
 - ii. In the Download Crosstab window, select the sheet you want to download.
 - iii. Click the **Download** button. The **Save/Save As** dialog box appears.
 - iv. Save the .CSV file to the location of your choice.
- d. Select the **PDF** option to download the dashboard/sheet in Portable Document Format (.PDF).
 - i. From the Download window, click **PDF**. The Download PDF window appears.
 - ii. In the Download PDF window, set up the .PDF file.
 - iii. Click the **Download** button. The **Save/Save As** dialog box appears.
 - iv. Save the .PDF file to the location of your choice.
- e. Select the **PowerPoint** option to download a sheet as a PowerPoint® presentation file in .PPTX format
 - i. From the Download window, click **PowerPoint**. The Download PowerPoint window appears.
 - ii. In the Download PowerPoint window, select the view/sheets you want to download.
 - iii. Click the **Download** button. The **Save/Save As** dialog box appears.
 - iv. Save the .PPTX file to the location of your choice.
- f. Select the **Tableau Workbook** option to download a workbook in Tableau® Workbook File (.TWB, an Extensible Markup Language (XML) file) format.

**NOTE**

Only the Administrator user has permissions to download Tableau workbooks.

- i. From the Download window, click **Tableau Workbook**. The Download Tableau Workbook window appears.
- ii. In the Download Tableau Workbook window, select the version of the workbook you want to download.
- iii. Click the **Download** button. The **Save/Save As** dialog box appears.
- iv. Save the .TWB file to the location of your choice.

5.8 Customer Assistance

If you need further information or have specific questions about the application (including self-service training options), contact your Digital Adoption Leader or Remote Trainer for assistance. In case of technical issues, log a service case.

A Dose Analytics Global Information

A.1 Link to DoseWatch

When selecting a dot or a line representing a study or a series, the corresponding DoseWatch study page will appear in another tab of the browser.

A.2 Metadata input for BMI, Age, Weight, Positioner Primary and Secondary angles ranges

Age, BMI, and Weight, Positioner Primary and Secondary angles ranges are according to the configuration done in the metadata file. By default, values are defined by the system but can be modified. Please contact your DoseWatch system administrator to proceed with the edition.

For DoseWatch system administrator, Metadata file can be found in <http://localhost:8085/md/ui/>, credentials for access are: user: configurator/password: imaging.insights. File name is: "range configuration".

- Click on the "View file" icon to view the file.
- Click on the "Modify file" for edition. Edit the "range-intervals" column following the instructions in the "comment" column. Save the edited file and Click on the "Upload file" icon for the changes to be considered by the system.

A.3 Standard Study Description - Maplex

Standard study description information is linked to the following two sources:

- DoseWatch Lexicon Mapping configuration.
- Maplex algorithm. Maplex is a functionality doing a mapping based on RPID's which links a protocol to the best matching RPID based on the core playbook. Maplex mapping inputs are the following:
 - Mapped by DoseWatch: if a protocol was already mapped in DoseWatch, Maplex will not edit it.
 - Mapped by Maplex: if a protocol was not yet mapped in DoseWatch, Maplex algorithm will identify the best matching RPID.
 - Unmapped protocol: if for any reason the user wishes to leave a protocol unmapped.



NOTE

Please contact your DoseWatch system administrator to proceed with the edition of this configuration.

For DoseWatch system administrator, Metadata file can be found in <http://localhost:8085/md/ui/>, credentials for access are: user: configurator/password: imaging.insights. File name is: "RPID mapping".

- Click on the "View file" icon to view the file.

- Click on the "Modify file" for edition. Edit the "rpid_user" column in the "Sheet1" to modify the mapping "rpid_default" and "metadata_status" columns will be automatically filled by maplex and the ETL run. Edit the "maplex_enabled" columns from "True" to "False" of the other sheets to disable the Maplex workflow. Save the edited file and Click on the "Upload file" icon for the changes to be considered by the system.

A.4 Action Filter

Action filter is available for the widgets containing the following icon ().

Action filter allows the user to filter the consulted dashboard by clicking into a specific widget item (trend line, bar, row) Please note that action filter will not be propagated to the other dashboards and that filtering options will not be reflected in the top right filter panel.

A.5 Level Configuration

Level configuration allows user to interact with the information hierarchization and granularity. Up to three items can be selected in the level configuration area. For each widget being impacted by the level configuration, expand and collapse actions are available when hovering the level heading column.

A.6 Zoom in/out in x axis when Containing Dates

Widgets where a representation of the Date information is available on the x-axis, a zoom in/out action is available when hovering over the area on the top left of the x-axis. This capability will then allow the user to visualize the data on a daily, weekly, monthly, quarterly, and yearly basis.

A.7 Metric Selector

Metric selector allows the user to interact with the information that should be represented in the consulted dashboard.

A.8 Specific Selectors

Specific selectors are available at a widget level allowing the user to interact with the data calculations, categorization, or display type for example. Some of those selectors are detailed below:

- Sort by # Studies/#Series or Alphabetically: user can select how the information that should be sorted based on the number of Studies/Series displayed or alphabetically.
- Chart Type: user can select the chart type representation in the specific widget.
- Percentile/Average: user can select a statistical metric to be overlayed in the specific chart.
- X axis: user can select which information should be displayed in the x axis in the specific chart.
- Legend: user can select how the information should be colored by in the specific chart.
- Show Total/Show by Device (Activity Overview).
- % Of Studies with filled (Data quality).
- Cumulative dose metric (Patient).

- Period of aggregation (Patient).
- DRL Type: user can select the DRL threshold type that should be considered for the specific chart calculations. DRL threshold types are configured in DoseWatch DRL module (Regional/National, Local and Alerts).
- % or #: user can select if the values should be represented in percentage or in absolute number.
- Highlight DRL Threshold: user can select which DRL Threshold should be highlighted in the corresponding widgets.

A.9 Filters

Filter panel contains the following filtering options:

- dropdown list: when displaying the dropdown list, users have access to the "All" selection or to a defined group of items. Please note that search option is available on the top of the dropdown list and that Apply button needs to be clicked for the action to be considered.
- slicers: user can interact with this filter type by moving the slicer or by hovering the min or max number and editing the value directly.
- value entry: user can directly enter the value of choice (example: date filter).
- Filter propagation: filters applied to a specific dashboard will be propagated to the other dashboards within the same workbook. Some exceptions are detailed below:
 - For CT Performance workbook, Outlier filter will not have an impact in Study Outliers or Series Outliers dashboards.
 - For CT Performance workbook, Pediatric dashboard filters will not have an impact in all the other dashboards and vice versa.
- Only relevant data: **Device, Study Protocol, Series Protocol, Series description, Model, DRL Name, Local Study Description, Standard Study description** filters are linked thanks to the "apply only relevant data" capability. Linked meaning that when a selection is done in one of those filters, values showed in the dropdown list of the others are automatically updated based on the previous selection. Example: if Revolution EVO device is selected, only Revolution EVO protocols will be displayed in the "Series Protocol" filter dropdown list.

A.10 Sorting

When hovering a specific column, user can click on the sorting icon to sort the data of the specific table based on the specific column information. Depending on the data type, sorting option will be alphabetically or numerically. Please note that if more than one level is displayed in the specific table, sorting will be done following the level granularity.

A.11 Tableau Top Left Actions

- Undo: Click the undo button to undo the previous action performed.
- Redo: Click the Redo button to redo the previous action performed.
- Replay: Click the Replay button to replay an animation.
- Revert: Click the Revert button to revert the view to the original published status or to the state the custom view was saved in.

- **Refresh:** Click the Refresh button to send a new query to the database to refresh the view. Please note that this action is only applicable if dashboard is Paused or if you are in a live connection.
- **Pause/Resume:** Click the Pause button to stop sending a query to the database and resume to execute it. Please note that this feature is useful when you want to apply multiple filters, but you do not want to wait for a refresh after each filter. In this case, you can Pause, select all your filters, click on Resume to execute only one query with all the filter configuration at the same time.

A.12 Creating Custom Views

Custom views with specific selectors and filter configuration can be saved. To create a custom view:

1. Filter the data, change sorting, configure specific selectors and/or make any other modifications as needed.
2. From the application toolbar, click View. The Custom Views dialog box appears.
3. In the **Name this view** field, type a name for the custom view.
4. Optionally, select the **Make it my default** check box if you want to make the custom view your default view.
5. Optionally, select the Make visible to others check box to make the custom view available to other users.
6. Click **Save**.

If you have selected the Make visible to others check box, click Share from the view's toolbar to share the custom view with other users. Please note that the customer's email server must be configured on the Tableau® server to share custom views.



NOTE

In case of software upgrade please note that existing views will not be saved.

A.13 Creating Alert

Alerts can be configured for the widgets having continuous numeric axis. To create an alert:

1. From the application toolbar, click **Alerts**. The Alerts pane appears on the right.
2. Select the continuous numeric axis of the widgets where the alert needs to be configured.
3. From the Alerts pane, click the **Create** button, The Create Alert window appears.
4. To Set up the alert. set the following:
 - a. **Condition** (drop-down list).
 - b. **Threshold** (text field).
 - c. **Subject** (text field). Please note that this field is pre-populated but editable.
 - d. **Send email frequency** (When the condition is true, send email drop-down list).
 - e. **Recipients** field. Start typing the username of a user to add the corresponding user to the alert email recipients.
 - f. **Email (text field)**. Please note that valid email address should be entered.

5. Click the **Create Alert** button. The alert is added to the Alerts pane. From the Alerts pane, the following actions are available for the alert by clicking the Actions drop-down menu:
 - a. **Remove me**
 - b. **Edit**
 - c. **Change owner**
 - d. **Delete**

A.14 Edit

Edit action allow the user to modify the dashboard. Only available for Administrator users.

A.15 Download

Entire workbooks, dashboards or sheets can be downloaded in different output formats. To download the content of the workbooks:

1. From the application toolbar, click Download. The Download window appears. Select the download format, following options are available:
 - a. Select the **Image** option to download the full dashboard view in Portable Network Graphics (.PNG) format.
 - b. From the Download window, click **Image**. The **Save/Save As** dialog box appears.
 - c. Save the .PNG file to the location of your choice.
2. Select the **Data** option to download a specific widget content in Comma Separated Values (.CSV) format. Please note that a specific widget needs to be previously selected for the Data option to be available and that only the data used to create the specific widget will be available.
 - a. From the Download window, click **Data**. The View Data window appears.
 - b. From the View Data window, click the **Download all rows as a text file link**. The Save/ Save As dialog box appears.
 - c. Save the .CSV file to the location of your choice.
3. Select the Crosstab option to download a selection of widgets as a text table/cross tabulation in .CSV format.
 - a. From the Download window, click Crosstab. The Download **Crosstab** window appears.
 - b. In the Download Crosstab window, select the widget/group of widgets you want to download and the file format (excel or csv).
 - c. Click the **Download** button. The **Save/Save As** dialog box appears.
 - d. Save the .CSV file to the location of your choice.
4. Select the **PDF** option to download the full dashboard view or selection of widgets in Portable Document Format (.PDF).
 - a. From the Download window, click PDF. The Download PDF window appears.
 - b. In the Download PDF window, set up the .PDF file.
 - c. Click the **Download** button. The **Save/Save As** dialog box appears.
 - d. Save the .PDF file to the location of your choice.

5. Select the **PowerPoint** option to download the full dashboard view or selection of widgets as a PowerPoint® presentation file in .PPTX format.
 - a. From the Download window, click **PowerPoint**. The Download PowerPoint window appears.
 - b. In the Download PowerPoint window, select the view/sheets you want to download.
 - c. Click the **Download** button. The **Save/Save As** dialog box appears.
 - d. Save the .PPTX file to the location of your choice.
6. Select the Tableau Workbook option to download a workbook in Tableau® Workbook File (.TWB, an Extensible Markup Language (XML) file) format. Please note that only the Administrator user has permission to download Tableau workbooks.
 - a. From the Download window, click **Tableau Workbook**. The Download Tableau Workbook window appears.
 - b. In the Download Tableau Workbook window, select the version of the workbook you want to download.
 - c. Click the **Download** button. The **Save/Save As** dialog box appears.
 - d. Save the .TWB file to the location of your choice.

A.16 Creating Comment

Comments can be created to allow users to note their comments and observations and share it with other users. To create a comment:

1. From the application toolbar, click Comments. The Comments pane appears on the right.
2. Type the comment.
3. Click **Post**.

A.17 Full Screen

Full screen mode allows to have a full screen view.

B Dose Analytics Dashboards Details

B.1 How are outliers identified in the box plot view?

Outliers' identification in the box plot view represented in the Distribution widget is using the following calculation:

- Lower limit: $P25 - 1.5 * (P75 - P25)$
- Upper limit: $P75 + 1.5 * (P75 - P25)$

Please note that this outlier visual representation is not linked to the Outliers dashboards configuration. However, the "include/remove outliers" filter, linked to the Outliers dashboards configuration will have an impact in the amount of data represented in the Distribution widgets.

B.2 How to Exclude/Keep only row option?

When hovering a selection of unique or multiple rows the Exclude/Keep only option will appear.

If the Exclude/Keep only action is selected for a row there will be an impact at the widget level only. Please note that this action will not be propagated to the other widgets or dashboards.

For multiple items selection drag-select or ctrl key actions are available.

Please note that Exclude/Keep only action will not be reflected in the top right filter panel.

B.3 How to Exclude/Keep only column option?

When hovering a selection of unique or multiple columns the Exclude/Keep only option will appear. This action will apply to all the widgets available in the consulted dashboard be applied at the widget level only. Please note that this action will not be propagated to the other widgets or dashboards and that it will not be reflected in the top right filter panel. For multiple items selection drag-select or ctrl key actions are available.

B.4 How to Exclude/Keep only dots option?

When hovering a selection after a drag-select of multiple dots the Exclude/Keep only option will appear. This action will be applied at the widget level only. Please note that this action will not be propagated to the other widgets or dashboards and that it will not be reflected in the top right filter panel. Please note that unique dot selection will lead to the "Link to DoseWatch" action.

B.5 How are these filters working?

Outliers filter:

Include/Remove Outliers filter action is related with the outlier calculation defined in the Study Outliers and Series Outliers dashboards. By default, Include Outliers filter option will be selected, when selecting Remove Outliers filter option, all applicable dashboards will only be representing non-outlier data. Please note that "Remove Study Outliers" option will only apply to Study level dashboards and that "Remove Series Outliers" option will only apply to Series level dashboards.

Diagnostic series & ratio filter:

Diagnostic series filter action is related with the Diagnostic Series ratio configuration. By default, all series data will be represented, when selecting Diagnostic Series filter option, only series which meets the following condition will be represented workbook. By default, Diagnostic Series ratio is 1.8, the user can change this value if applicable.

B.6 How to configure metadata for Cumulative x-ray duration?

Cumulative X-ray duration calculation is done based on tube installation or change date information according to the configuration done in the metadata file. Please note that by default this widget will be empty, tube installation or change date needs to be entered in the metadata file for the widget to represent cumulative x-ray duration. Please contact your DoseWatch system administrator to proceed with the metadata file edition.

For DoseWatch system administrator: Metadata file can be found in <http://localhost:8085/md/ui/>, credentials for access are: user: configurator/password: imaging.insights. File name is: "Tube change date configuration".

- Click on the "View file" icon to view the file.
- Click on the "Modify file" for edition. Edit the "last_tube_change_date" column following the format YYYY-MM-DD. Save the edited file and Click on the "Upload file" icon for the changes to be considered by the system.

B.7 CT Performance - Study Outliers

This section determines Outliers based on the "Metric" selected through upper and lower limit thresholds typed, automatically or not typed value. All values calibrated here are applied to all charts on this screen. First, a selection on which dose metric the outliers shall be based on needs to be done. As an example, for CT modality, following options are available (Metric):

- Total DLP(mGy.cm)
- Total Number of Irradiation Events
- Max CTDIvol (mGy)
- Total Effective Dose (mSv)
- Revised Total DLP (mGy.cm)
- Total Effective Dose (from Organ Dose) (mSv)

Second, a selection on the descriptor to group the information needs to be done. As an example, for CT modality, following options are available (Outliers Basis):

- Local Study Description
- Standard Study Description
- Study Protocol
- Series Protocol
- Anatomical Region
- Study DRL Name

Following outlier's limits options are available:

- Lower and Upper outliers
- Lower outliers only
- Upper outliers only

Following outlier's calculations options are available:

- None: No limit defined.
- Auto: If auto input is selected, the outlier limit(s) will be calculated automatically for each item within the selected descriptor (if Study Protocol is selected, the outlier limits will be calculated for each Study protocol). The formula over which outliers are calculated are the following:

- Lower limit: $P25 - \text{scale} * IQR$
- Upper limit: $P75 + \text{scale} * IQR$

By default, the scale is set to 1.5, but I can change it to any value I want between 1 and 3

- Manual: If manual input is selected for an upper and / or lower limit, those limits will apply to each item within the selected descriptor (if Study Protocol is selected, the outliers' limits will be the same for each Study Protocol).

The outlier calculation selection will be considered if the Remove Study Outliers filter option is selected in other workbook dashboards.

B.8 CT Performance - Series Outliers

This section determines Outliers based on the "Metric" selected through upper and lower limit thresholds typed, automatically or not typed value. All values are calibrated here are applied to all charts on this screen. First, a selection on which dose metric the outliers shall be based on needs to be done. As an example, for CT modality, following options are available (Metric):

- CTDI vol (mGy)
- DLP(mGy.cm)
- SSDE (mGy)
- Effective Dose (mSv)
- Revised CTDIvol (mGy)
- Revised DLP (mGy.cm)
- Effective Dose (from Organ Dose) (mSv)

Second, a selection on the descriptor to group the information needs to be done, following options are available (Outliers Basis):

- Local Study Description
- Standard Study Description
- Series Protocol
- Series Description
- Series DRL Name
- Anatomical Region

Following outlier's limits options are available:

- Lower and Upper outliers
- Lower outliers only
- Upper outliers only

Following outlier's calculations options are available:

- None: No limit defined.
- Auto: If auto input is selected, the outlier limit(s) will be calculated automatically for each item within the selected descriptor (if Study Protocol is selected, the outlier limits will be calculated for each Study protocol). The formula over which outliers are calculated are the following:

- Lower limit: $P25 - \text{scale} * IQR$
- Upper limit: $P75 + \text{scale} * IQR$

By default, the scale is set to 1.5, but I can change it to any value I want between 1 and 3

- Manual: If manual input is selected for an upper and / or lower limit, those limits will apply to each item within the selected descriptor (if Study Protocol is selected, the outlier's limits will be the same for each Study Protocol).

The outlier calculation selection will be considered if the Remove Series Outliers filter option is selected in other dashboard workbook dashboards.

B.9 Additional Details

- **List of patients when clicking into the bar chart allowing to select a patient to go to DW patient dose history tab:**

When clicking into a specific bar, the list of patients being represented will appear. When clicking into a specific patient ID, the corresponding DoseWatch Patient Dose History page will appear in another tab of the browser.

- **Link to DoseWatch:**

When clicking into a dot representing a patient, the corresponding DoseWatch Patient Dose History page will appear in another tab of the browser.

- **Difference between alert and threshold exceed:**

DoseWatch alert management can be done based on multiple study or series KPIs. Threshold exceed information will represent the number of study or series KPIs where the alert threshold was exceeded. Alert information will represent the number of studies where at least one threshold was exceeded.

- **DRL id (not available in DoseWatch but available in DA):**

In DoseWatch DRL module configuration no restriction exists regarding the DRL's Name duplication. To differentiate properly the data in the DRL analysis, the DRL ID is represented in brackets near to the DRL Name information.

- **Percentile/Average configurator:**

A statistical calculation needs to be provided to the system for the Statistics table to be filled. Columns representing a # will not be impacted by this calculation choice as they are representing cumulated values.

- **Annotations selector:**

Annotation selector allows the user to choose how the Annotation information should be displayed:

- Code + Label: for example, DW01 High BMI

- Code only: for example, DW01
- Label only: for example, BMI
- **Internal links:**

When clicking on any set of data contained in the Annotations overview widget an action filter will be applied and user will be automatically redirected to the Alert details dashboard to have a detailed view on the data selection.
- **Correlation, trend:**

When hovering the top right area of a legend the "Highlight selected items" option will appear. A legend item or group of items should be previously selected, once "Highlight selected items" is selected, those items will appear highlighted in the corresponding widget while the others will be grayed out. To come back to the initial status, all items highlighted, unselect "Highlight selected items".



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